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**4th Sem / Auto, Mech, Prod, T&D, GE, CAD/CAM,
CNC, Metallurgy, Found. & Forg., Adv. Manuf. Tech.,
Mech Engg (Fabrication Tech) , Mech Engg (CAD /
CAM dsgn & Robotics)**

Subject:-Materials and Metallurgy / Mat. Sci.

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 How is the creep strength of ceramic when compared to other materials? (CO1)

- a) Low b) High
c) Excellent d) Zero

Q.2 Which of these is not a function of alloy steels?
(CO1)

- a) Improves ductility
- b) Improves machinability
- c) Increases strength
- d) Reduces cost

Q.3 Which of the following is not a classification of aluminum alloys? (CO3)

- a) Cast alloys b) Heat-treatable alloys
c) Wrought alloys d) Crucible alloys

Q.4 Coordination number of HCP and FCC lattices respectively are: (CO2)

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- a) 12,12 b) 4,4
c) 12,8 d) 8,8
- Q.5 Allotropes differ in which of the following properties: (CO2)
a) Atomic number b) Atomic Mass
c) Crystal Structure d) Electronegativity
- Q.6 Which of the following factors does not influence the variety and quality of metal? (CO2)
a) Rate of heating and cooling
b) Quenching medium
c) Furnace
d) Grain size
- Q.7 _____ is used to predict quenching reactions in steels. (CO4)
a) Isothermal transformation diagram
b) Iron-iron carbide equilibrium diagram
c) Continuous cooling transformation diagram
d) Lagorithm scale
- Q.8 Hot-quenching of eutectoid steels in austenitic condition results in formation of _____ (CO4)
a) Pearlite b) Bainite
c) Ferrite d) Cementite
- Q.9 Which of the following factors do not affect the critical cooling rate? (CO4)
a) Chemical Composition
b) Hardening temperature
c) Number or nature of grains
d) Purity of steel

- Q.10 The fabrication cost is _____ for plastics. (CO6)
- a) High b) Low
- c) Moderate d) Very high

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

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|------|---|-------|
| Q.11 | Define crystal structure. | (CO1) |
| Q.12 | Define atomic packing factor. | (CO2) |
| Q.13 | Define creep. | (CO3) |
| Q.14 | Define unit cell. | (CO1) |
| Q.15 | Define allotropy. | (CO4) |
| Q.16 | What is aim of heat treatment? | (CO3) |
| Q.17 | Carburizing is not surface hardening process (True/False) | (CO4) |
| Q.18 | Copper is a ductile material.(True/False) | (CO5) |
| Q.19 | Define Plastics. | (CO6) |
| Q.20 | Name any two types of Rubber. | (CO7) |

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What is the difference between metal and non-metals. (CO1)
- Q.22 Write short note on classification of materials. (CO1)
- Q.23 Differentiate slip and twinning. (CO2)
- Q.24 Explain the effect of grain size on the properties of materials. (CO2)
- Q.25 Discuss heat treatment and its objectives. (CO3)

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- Q.26 Write the effects of alloying tungsten and chromium in steel. (CO4)
- Q.27 Write at least five applications of high carbon steel. (CO5)
- Q.28 Explain the various grades of stainless steel. (CO5)
- Q.29 Show flow diagram for production of iron. (CO4)
- Q.30 Write the process of manufacturing of stainless steel. (CO5)
- Q.31 Explain martensitic transformation. (CO4)
- Q.32 Define the method of annealing. State its purpose. (CO3)
- Q.33 Describe Schaeffler diagram. (CO5)
- Q.34 What should be the desired properties of bearing materials. (CO7)
- Q.35 Write the five properties of asbestos. (CO8)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 What is creep. Explain different types of creep. Draw creep curve also. (CO1)
- Q.37 Define cast iron? Give in detail the types, their properties and uses. (CO8)
- Q.38 Explain Iron-carbon diagram in detail. (Co4)